Department of BSC.CSIT

Lab Manual

Academic Year: 2083

Microprocessor

Prepared By:

Er. Nirmal Thapa

Butwal Multiple Campus

(Sample Cover page, please use your own university/college name& department for your lab report submission)

Butwal Multiple Campus

Tribhuvan University

Butwal, Rupandehi, Nepal

Lab Report on Microprocessor

Faculty of B.Sc CSIT

Tribhuvan University

Kritipur, Nepal

<u>Submitted By</u>

Name: Your Name

Roll No: xxx

<u>Submitted To</u>

Butwal Multiple Campus

Department of BSC CSIT

Butwal, Rupandehi Nepal

-------------	-----------------------	-----------------------
Signature	Internal Examiner	External Examiner

This Lab Manual is prepared to help the students with their practical understanding and development of programming skills and knowledge on Microprocessor, and may be used as a base reference during the lab/practical classes.
Students have to prepare Lab report of all lab done in class. At the end of the semester, students should compile all the Lab Exercise reports into a single report and submit during the end semester sessional examination.

Sample of Lab report for your reference, “How to write a complete lab report” is shown in this manual.

The lab report to be submitted should (only for program) include at least the following topics:

1. Cover page
2. Title
3. Objective(s) / AIM
4. Algorithm
5. Program
6. Output
7. Discussion & Conclusion.

EXERCISE NO. 1

ADDITION OF TWO 8 BIT NUMBERS

OBJECTIVE:

To perform the addition of two 8 bit numbers using 8085 Kit (MPS 85-3) and also run the program in 8085 Simulator.

ALGORITHM

1) Start the program by loading the first data into Accumulator.
2) Move the data to a register (B register).
3) Get the second data and load into Accumulator.
4) Add the two register contents.
5) Check for carry.
6) Store the value of sum and carry in memory location.
7) Terminate the program.

SOURCE CODE

LOCATION	MNEMONICS	OPCODE	COMMENTS
2000	MVI C, 00	0E 00	Initialize register C to 00 (used to store carry)
2002	LDA 4150	3A 50 41	Load value from memory address 4150 into Accumulator
2005	MOV B, A	47	Move content of Accumulator to register B
2006	LDA 4151	3A 51 41	Load value from memory address 4151 into Accumulator
2009	ADD B	80	Add contents of register B to Accumulator
200A	JNC LOOP	D2 0F 20	Jump to LOOP if no carry is generated
200D	INR C	0C	Increment register C (used to count carry)
200E	**LOOP:**		Label for jump location
200F	STA 4152	32 52 41	Store the result (SUM) from Accumulator to memory address 4152
2012	MOV A, C	79	Move the content of register C to Accumulator
2013	STA 4153	32 53 41	Store the carry value from Accumulator to memory address 4153

SAMPLE INPUT & OUTPUT

Memory Address	Value	Meaning

4152	**07** (decimal 7)	**SUM** (lower 8 bits only)
4153	**01** (decimal 1)	**CARRY** (indicates overflow beyond 8 bits)

RESULT

Thus, the program to add two 8-bit numbers using 8085 kit (MPS 85-3) was executed successfully.

Lab Exercises (Programs List):

1. Write 8085 assembly language program for multiplication of two hexadecimal numbers.
2. Write 8085 assembly language program for division of two hexadecimal numbers.
3. Write an ALP program to search the largest number in a given array using 8085 microprocessor. Store the result at the end of the array.
4. Write an ALP program to sort data in descending order in an array using 8085 microprocessor.
5. Write an ALP program for addition of two 16-bit numbers.
6. Write 8086 assembly language program for multiplication of two hexadecimal numbers.
7. WALP to search the character ‘e’ in a string “computer science” and count the number of repetitions in a string using 8086 microprocessor.
8. WALP to convert the given string ‘csit’ into uppercase alphabet using 8086 microprocessor.
9. WALP to reverse the given string using 8086 microprocessor.
10. WALP to find the factorial of a given number using 8086 microprocessor.